



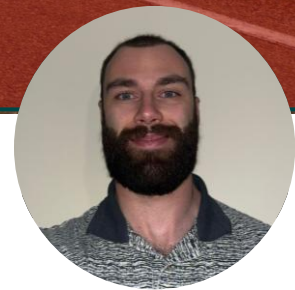
Amplifying Productivity in the Prairies



Christian Arteaga



Mary Jane Watkins



Duncan Morrison

How can PrairiesCan raise productivity and competitiveness across the Prairies?

Executive Summary

Pain points

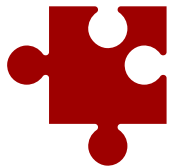


Stagnant Productivity Across
the Prairies

Low Business Investment

Constrained Technology
Adoption

Strategy



The Prairie Growth Engine

Flexible Scale-Up
Support



Technology adoption



Policy
Recommendations



Impacts



Improved productivity
performance

Increased confidence in
adopting new technologies

Higher awareness of available
support programs

Analysis

Preliminary Considerations

Productivity

Defining Productivity

1. Multifactor productivity (MFP), often referred to as total factor productivity (TFP), measures economic output per unit of combined inputs (such as labour, capital, materials, and energy)
2. It captures the portion of output growth that is not explained by increases in individual inputs

Measurement Challenges

1. Canada's economy is predominantly service-based, and service productivity is difficult to define and to capture
2. In capital-intensive industries, labour productivity figures are mechanically high as an artifact of how they are calculated: more capital per worker inflates the output-to-labour ratio

Examples

Challenge #1 | Healthcare:

A doctor who sees fewer patients in a day may actually be delivering more valuable care if those cases are more complex or require more time

Challenge #2 | Alberta's oil sands:

Large investments in machinery, equipment, and extraction infrastructure allow each worker to produce a high level of output, so the output-per-worker measure is inflated by capital

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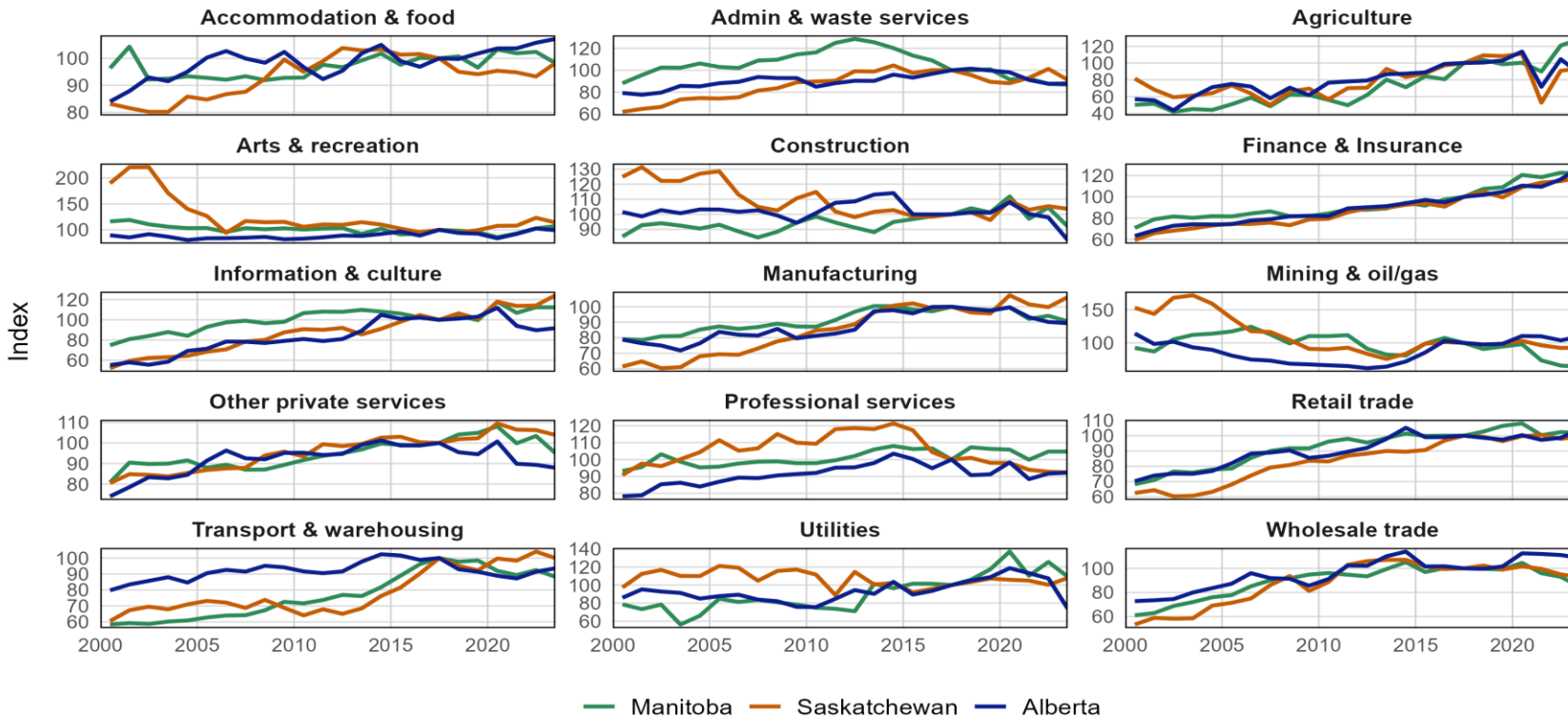


Productivity Analysis

Labour Productivity

Labour Productivity Across the Prairies

Historical trend by industry and province from 2000 to 2023 (Index, 2017=100)



Source: Statistics Canada, Table 36-10-0480-01

Key Findings:

- Most sectors are benefiting from economies of scale in early stages
- More integrated supply chains improve efficiency
- Mining, Oil and Gas sector is highly sensitive to global commodity prices, making productivity more dependent on external market conditions.

Key Takeaway: Most industries show an overall upward trend in labour productivity over time.

Productivity Analysis

Multifactor Productivity

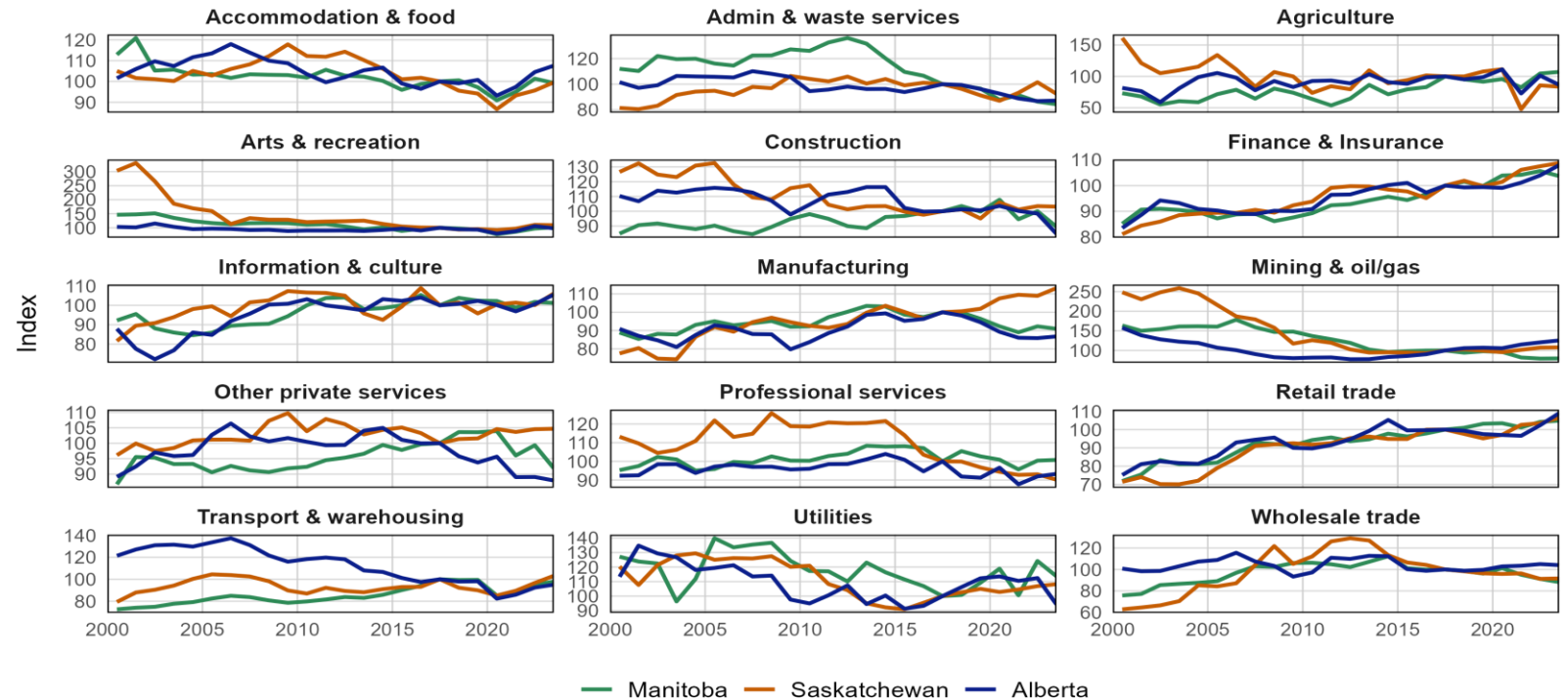
Key Findings:

Possible reasons why productivity has not increased as much as in other economies

- Weaker investment in advanced capital
- Slower diffusion of new technologies and management practices across firms

Multifactor Productivity Across the Prairies

Historical trend by industry and province from 2000 to 2023 (Index, 2017=100)



Source: Statistics Canada, Table 36-10-0480-01

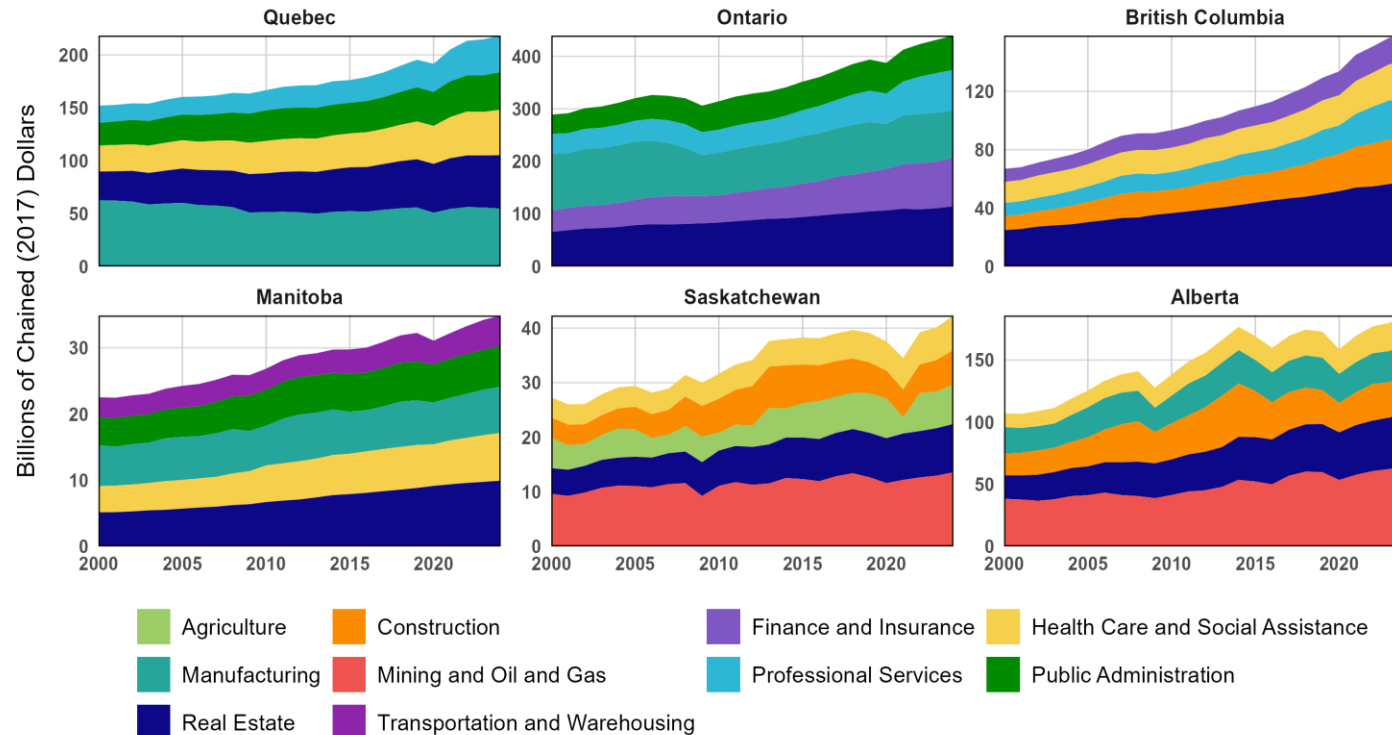
Key Takeaway: Recent gains in output have been driven by greater input use or only limited changes in how labour and capital are combined

Productivity Analysis

Real Gross Domestic Product

Top 5 Industries by Real GDP in Each Province

Historical trend for the five largest industries in 2024



Source: Statistics Canada Table 36-10-0711-01

Key Findings:

1.

British Columbia and Ontario:
Housing markets, urban development, and related service activity as main economic drivers

2.

Quebec, Ontario and Manitoba:
Large manufacturing share

3.

Alberta and Saskatchewan:
Stronger dependence on resource extraction and greater exposure to commodity price cycles.

Key Takeaway: Clear differences in provincial economic structure determining their specialization and output

Opportunities

Opportunity #1

Agriculture and Agri-food Sector

Agriculture in Canada stands as a success story

Goals

1. In 1921, approximately one third of Canadian jobs were in farming. Today, that figure is roughly 1.5%
2. Cultivated farmland and average farm size have both grown, allowing higher output with fewer workers through technological improvements.

What we can improve on

1. Agricultural programs are chronically undersubscribed (31% of eligible producers were enrolled in AgriStability, a program designed to insure farmers against abnormally poor yield years - 2020)
2. These programs represent substantial funds already earmarked for agricultural productivity; the barrier is adoption.

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Goals



Goal #1:

Increase program uptake at modest cost through targeted outreach to agricultural firms and more robust matching tools



Goal #2:

Amplify existing initiatives with complementary programs designed to lower entry barriers and increase competition

Opportunity #1

Multifactor Agricultural Productivity

Multifactor Agricultural Productivity Across the Prairies

Historical trend by industry and province from 2000 to 2023 (Index, 2017=100)



Source: Statistics Canada, Table 36-10-0480-01

To improve agricultural productivity...

Recommendation #1

Improving Awareness and Accessibility of Undersubscribed Programs

Ex: Enhance the already extant Agpal with more robust AI-assisted matching and standing alert systems to improve program uptake

Recommendation #2

Offering Complementary Prairies-specific Incentives Alongside Existing Federal Initiatives

Ex: Offer complementary regional grants or low-interest loans covering startup costs associated with federal greenhouse program

Recommendation #3

Developing Standing Capacity to Respond to Time-Sensitive Trade Opportunities

Ex: Create and maintain capacity to respond quickly to time-sensitive opportunities, such as recent Chinese trade agreements

Opportunity #2

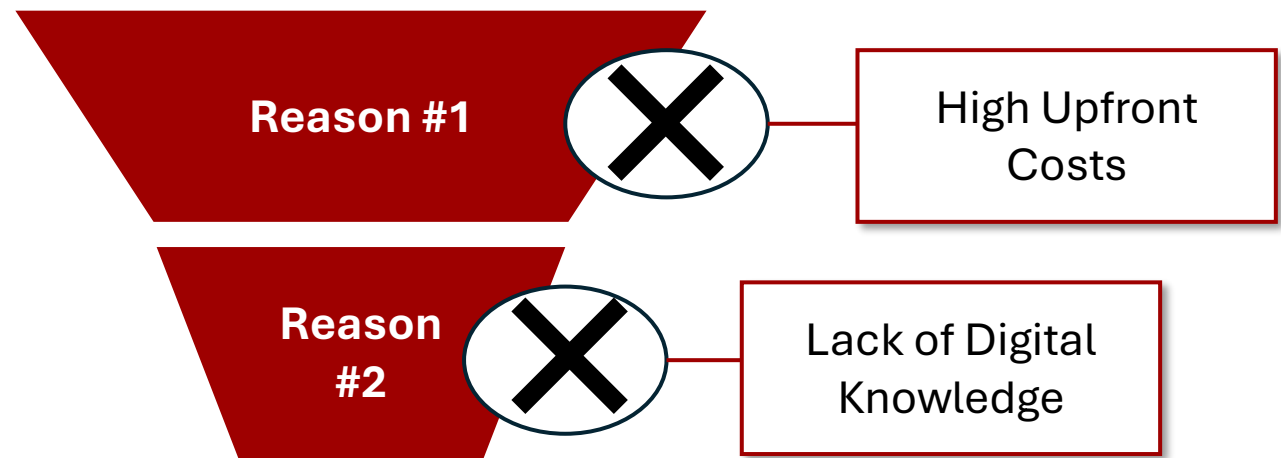
Business investment

Reasons why Canada has fallen behind in economic growth...

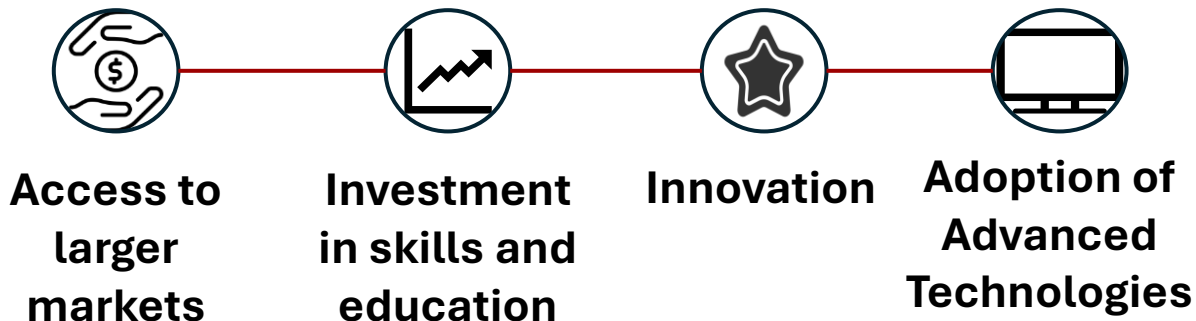


- Weak Capital Investment
- Smaller Firms

If there are gains to productivity that can be achieved from technology adoption, why do Canadian firms not invest in technology?



How larger firms are more productive:



In times of uncertainty, many firms favour short-term horizons, often focusing on maintaining existing competitive advantages rather than creating new ones

Many firms also believed that there are barriers to entry in their industry; thus, they were not induced by competition to engage in business investment

Opportunity #2

PrairiesCan's Current Measures

PrairiesCan already has two programs

1. **Business Scale-Up productivity (BSP):** It supports high growth businesses in the Prairies seeking to accelerate growth through productivity improvement, business-scale up, or technology commercialization, with preference given to SMEs.
2. **Regional Quantum Initiative (RQI):** program supports Albertan for-profit businesses and not-for-profit organizations engaged in quantum technology



Programs

Business Scale-Up Productivity

- Interest-free
- Repayable funding of \$200,000 to \$5 million per project
- Covering up to 50% of eligible costs

Regional Quantum Initiative

- Interest-free
- \$360 million over 7 years in dedicated federal funding
- Repayable funding for up to 50% of eligible costs
- Eligible not-for-profits can receive non-repayable funding for up to 90% of eligible costs

Opportunity #2

Research and Development

R&D as a Growth Catalyst

Why R&D Matters:

R&D can create both new technologies and bolster competition, both of which have positive spillovers

How Incentives Can Crowd In Investment:

Providing incentives for innovation in a way that reduces risk to business can attract outside capital and highly-qualified individuals (HQIs), which will also increase productivity

Evidence of Large-Scale Innovation Potential:

The sparsely settled land is suitable for building high-tech facilities, and recently there have been plans to build data centres precisely to capitalize on this opportunity

Building them shows land use potential for businesses that wish to engage in R&D at a large scale and suggests producer interest in innovation



Rocky View County Data Centre Project

Opportunity #3

Technology Adoption

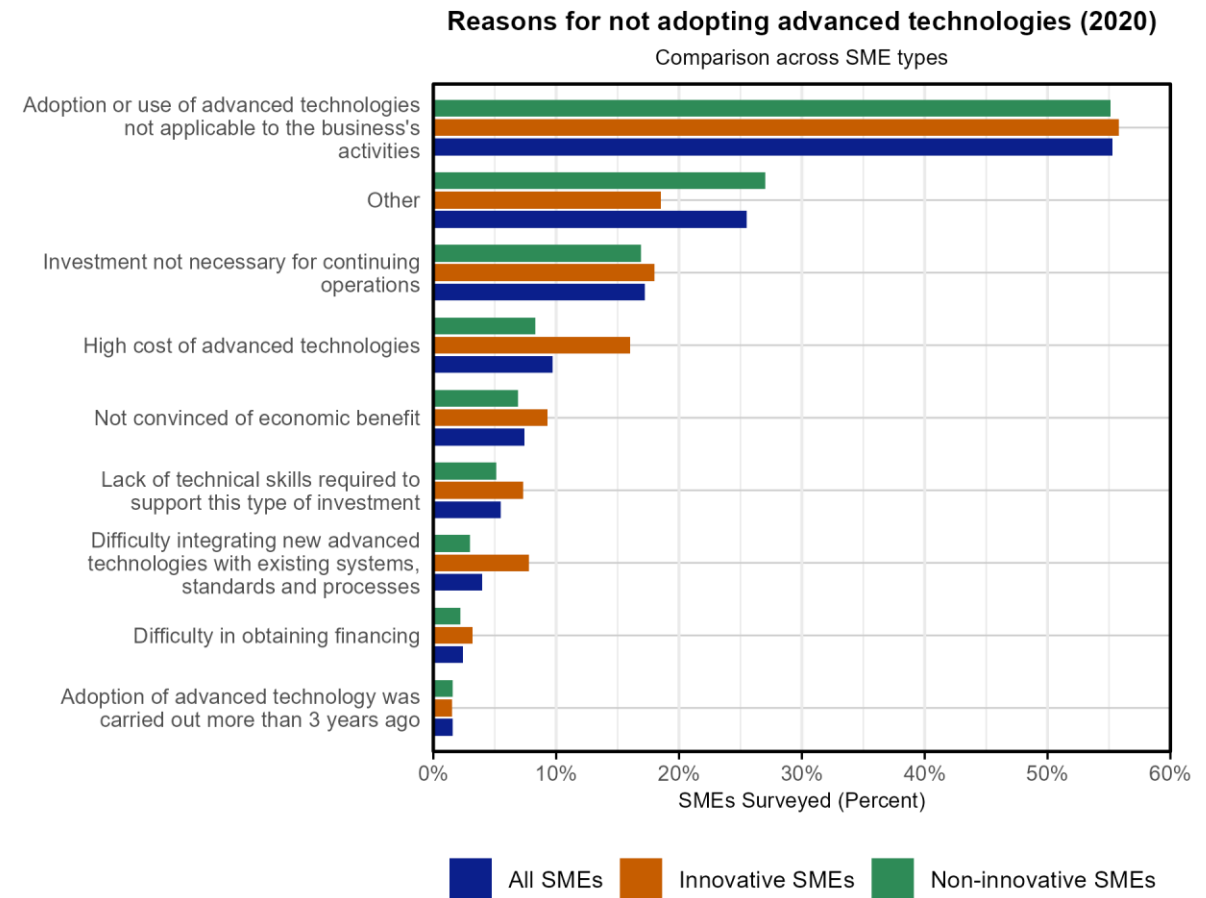
Benefits of Technology Adoption

1. Allows firms to decrease their costs from increased productivity
2. Can help small and medium-sized manufacturers improve product quality, reduce downtime, and strengthen demand forecasting through the use of sensors, analytics, and artificial intelligence

Questions about Technology Adoption

1. Will this actually improve productivity?
2. How much will it cost to adopt and maintain?
3. Is this technology suitable for a business our size?

Main Reasons Why They are not Implemented



Opportunity #3

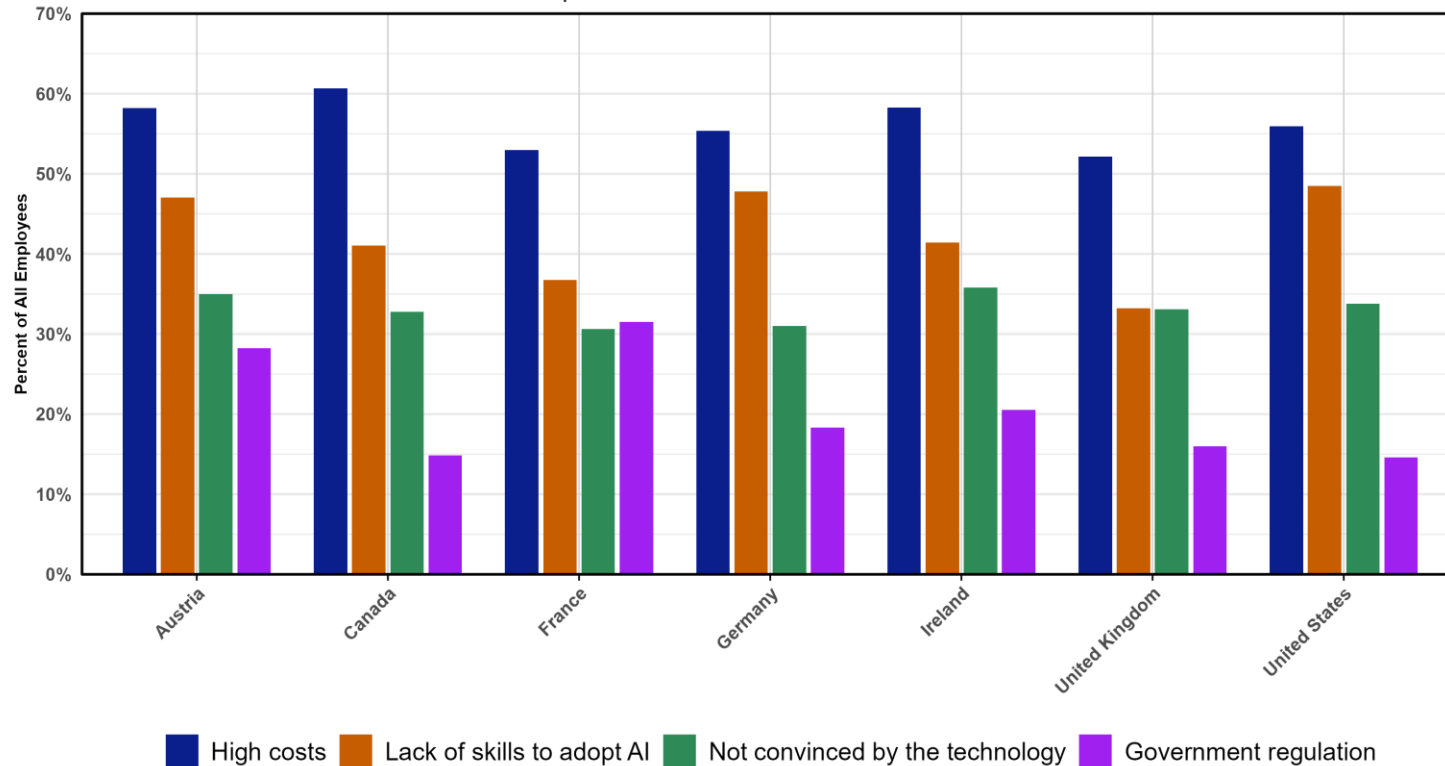
Technology Adoption

Problem

Developing AI models and storing and analyzing large amounts of data can be expensive

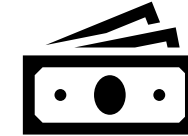
Barriers to AI Adoption by Country (2022)

Reported barriers across selected countries



Source: Lane, M., M. Williams and S. Broecke (2023)

Common Barrier



High Costs



Lack of Skills



Skepticism About
Technology Benefits

Opportunity #4

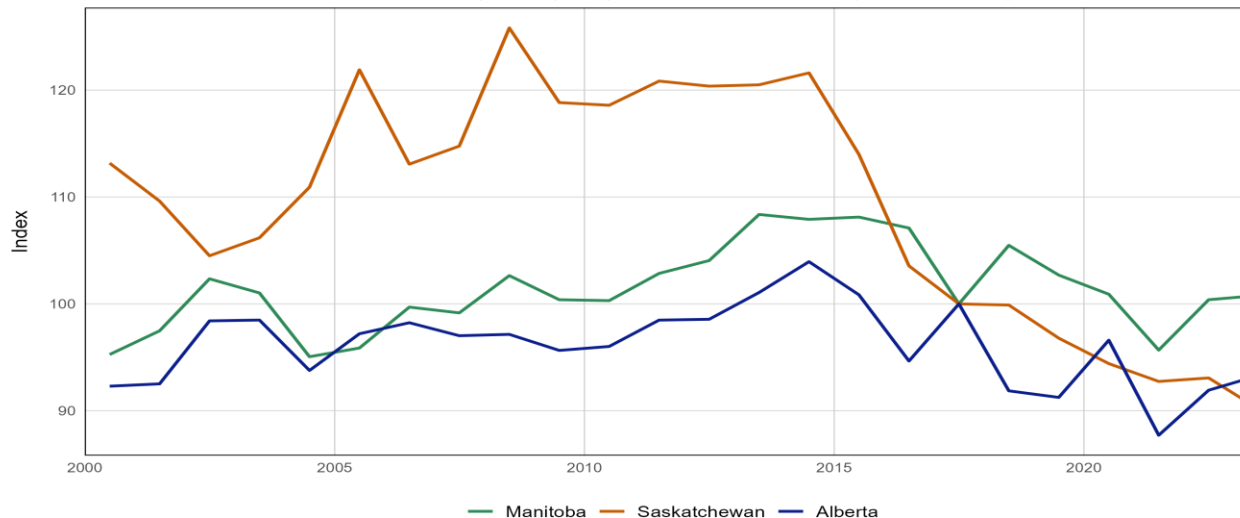
Management and Professional Services

Managerial Challenges for SMEs

1. SMEs often struggle to accumulate the specialized human capital needed to improve management and scale effectively, which limits their ability to compete with larger firms.
2. As large businesses continue to attract and retain this talent, SMEs face greater barriers to growth and remain under stronger competitive pressure

Multifactor Professional Services Productivity Across the Prairies

Historical trend by industry and province from 2000 to 2023 (Index, 2017=100)



Source: Statistics Canada, Table 36-10-0480-01

Recommendation



Recommendation:

Investigate programming that subsidizes management and business fundamentals training initiatives, with a specific focus on small to midsize firms

Goals

- Firms with good ideas should grow and attract talent on their own merits
- Reduce the structural barrier that currently prevents specific categories of firms from engaging in such competition



The Prairie Growth Engine

1

Flexible Scale-Up Support

Expand adaptable funding and support tools that help firms grow based on their size, stage, and sector-specific needs

2

Technology Adoption

Help firms adopt productivity-enhancing technologies by easing financial barriers

3

Policy Recommendations

Refine programs and incentives so they better address firm needs and generate stronger productivity outcomes

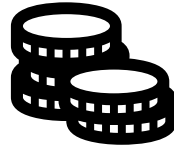
Flexible Scale-Up Support

Research and Development

BSP Program Improvement

1.

Decrease the minimum loan amount offered in the program from \$200,000 to any amount that a business needs



2.

For low value loans, allow for a loan that covers the full cost of the technology rather than the standard 50 percent



3.

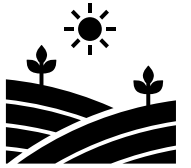
Make Prairie businesses aware of opportunities available to them where funding has already been approved



RQI Program Expansion

1.

Assist SMEs that wish to engage in R&D in any field in the Prairies



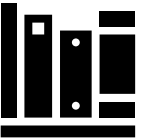
2.

Transferring some risk away from firms



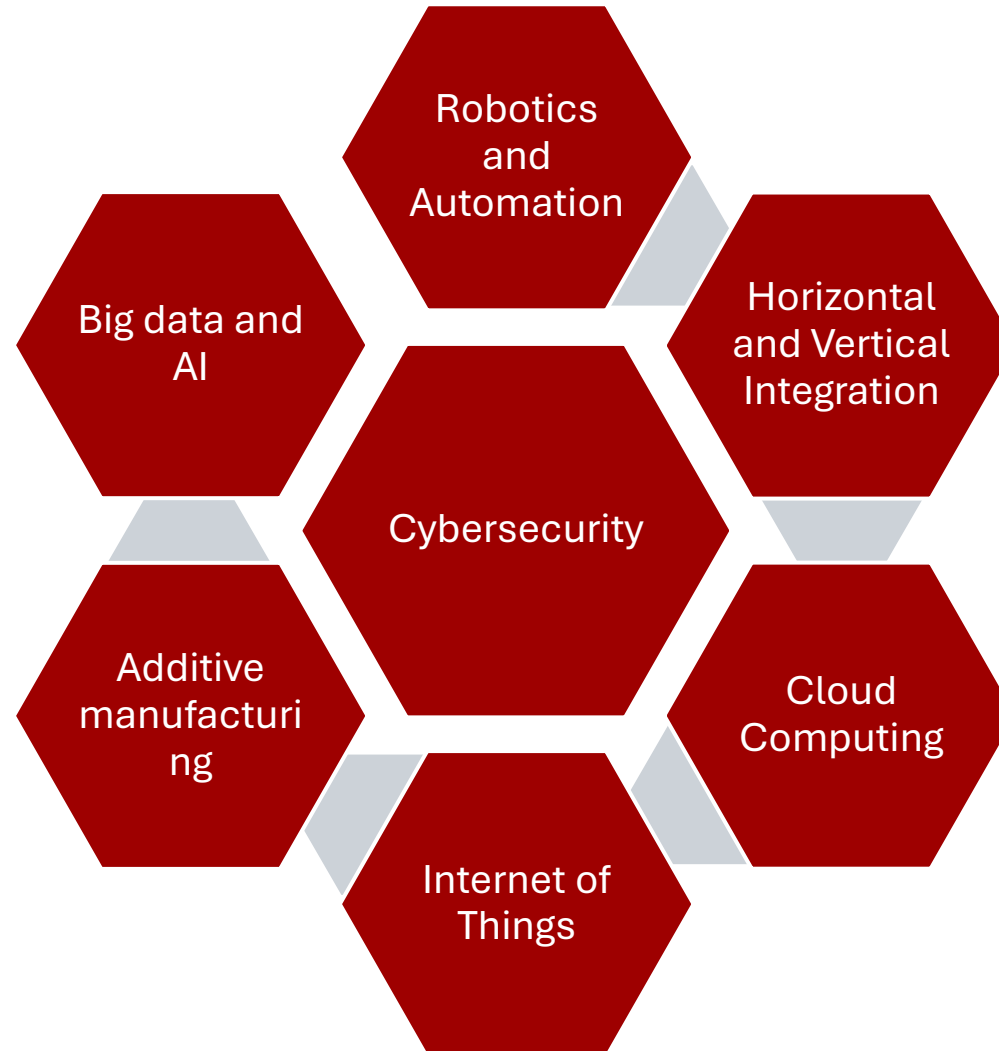
3.

Build shared knowledge that raises the chances of future breakthroughs



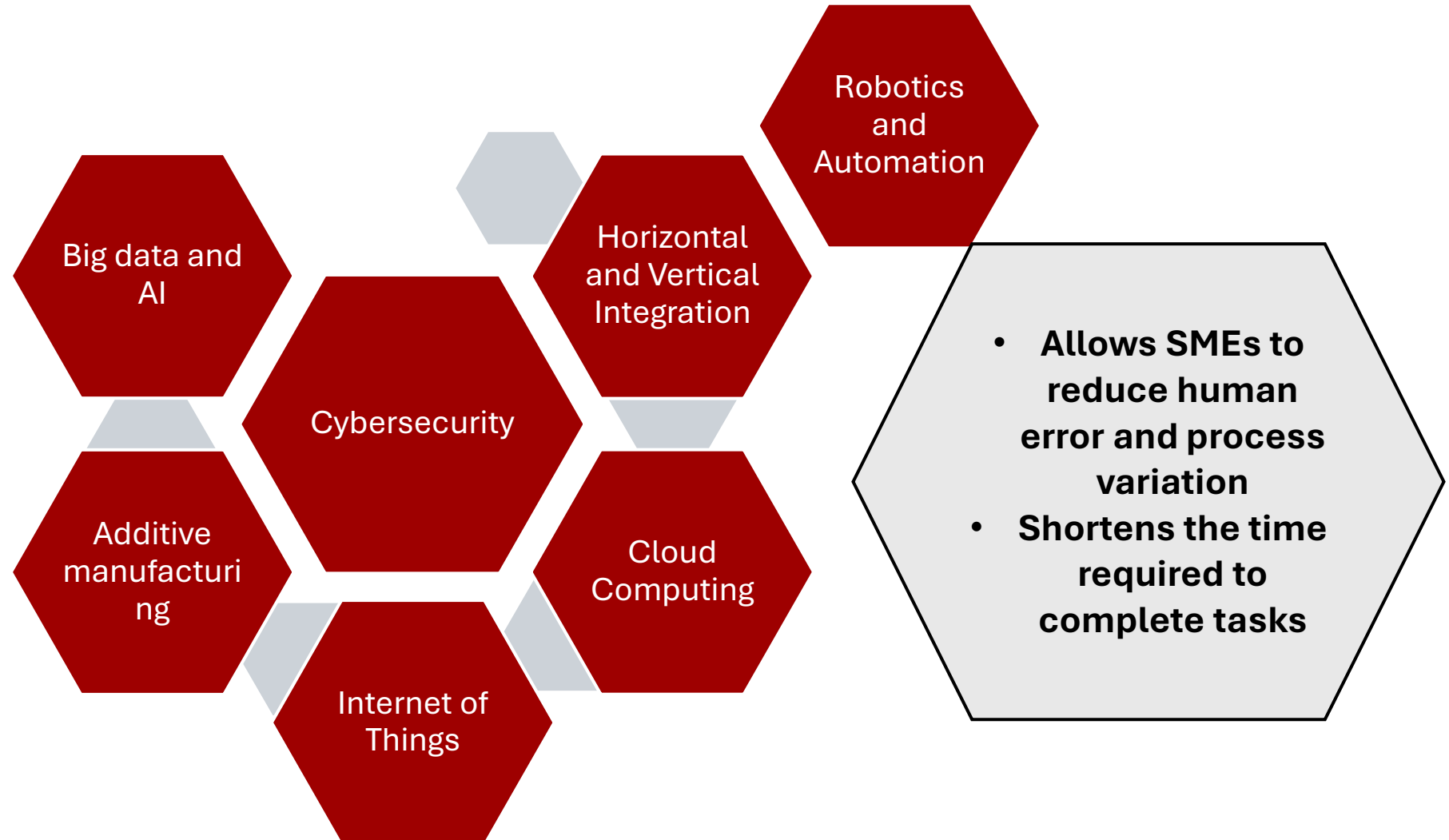
Technology Adoption

Key Technologies Transforming Industrial Production



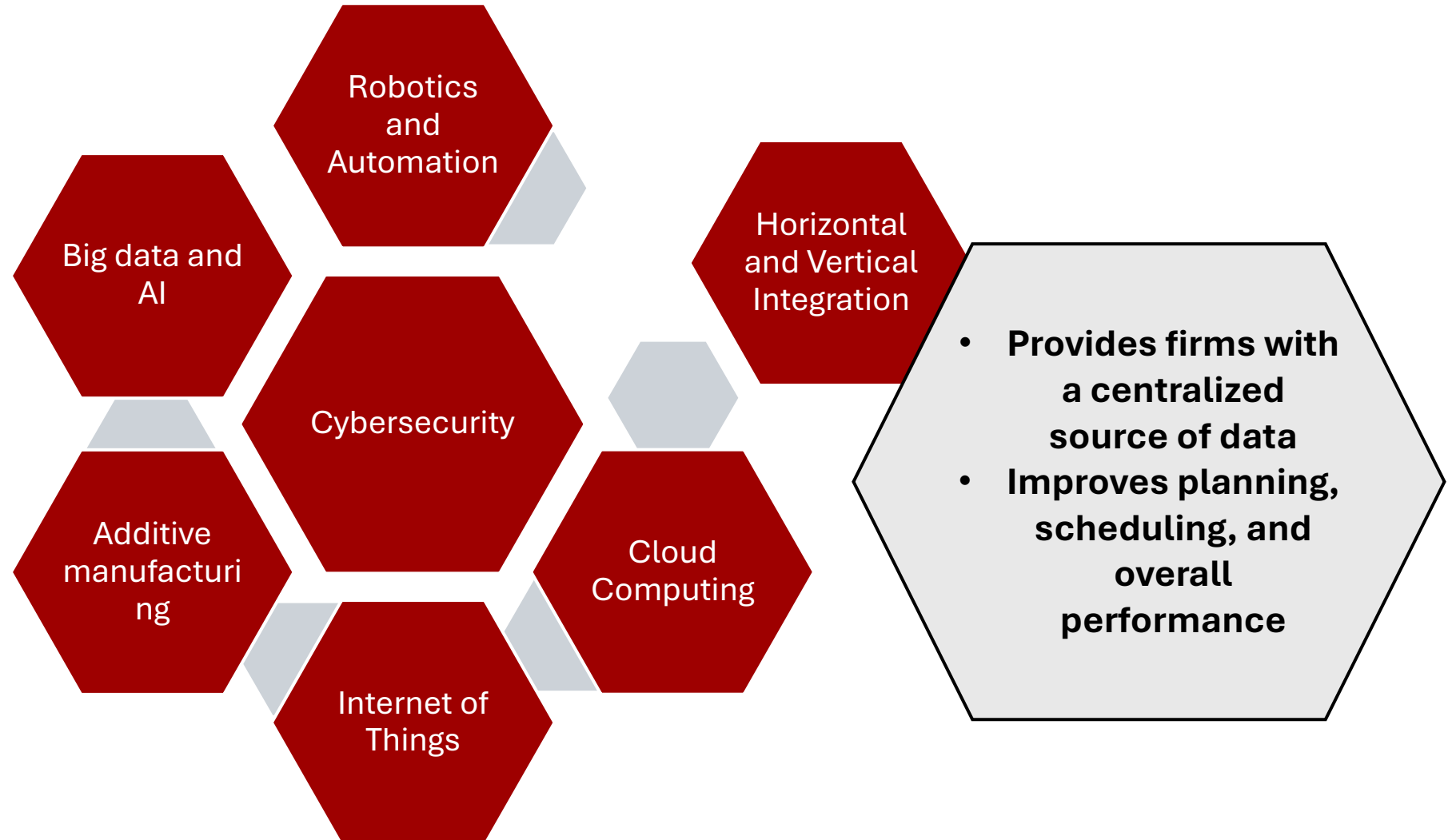
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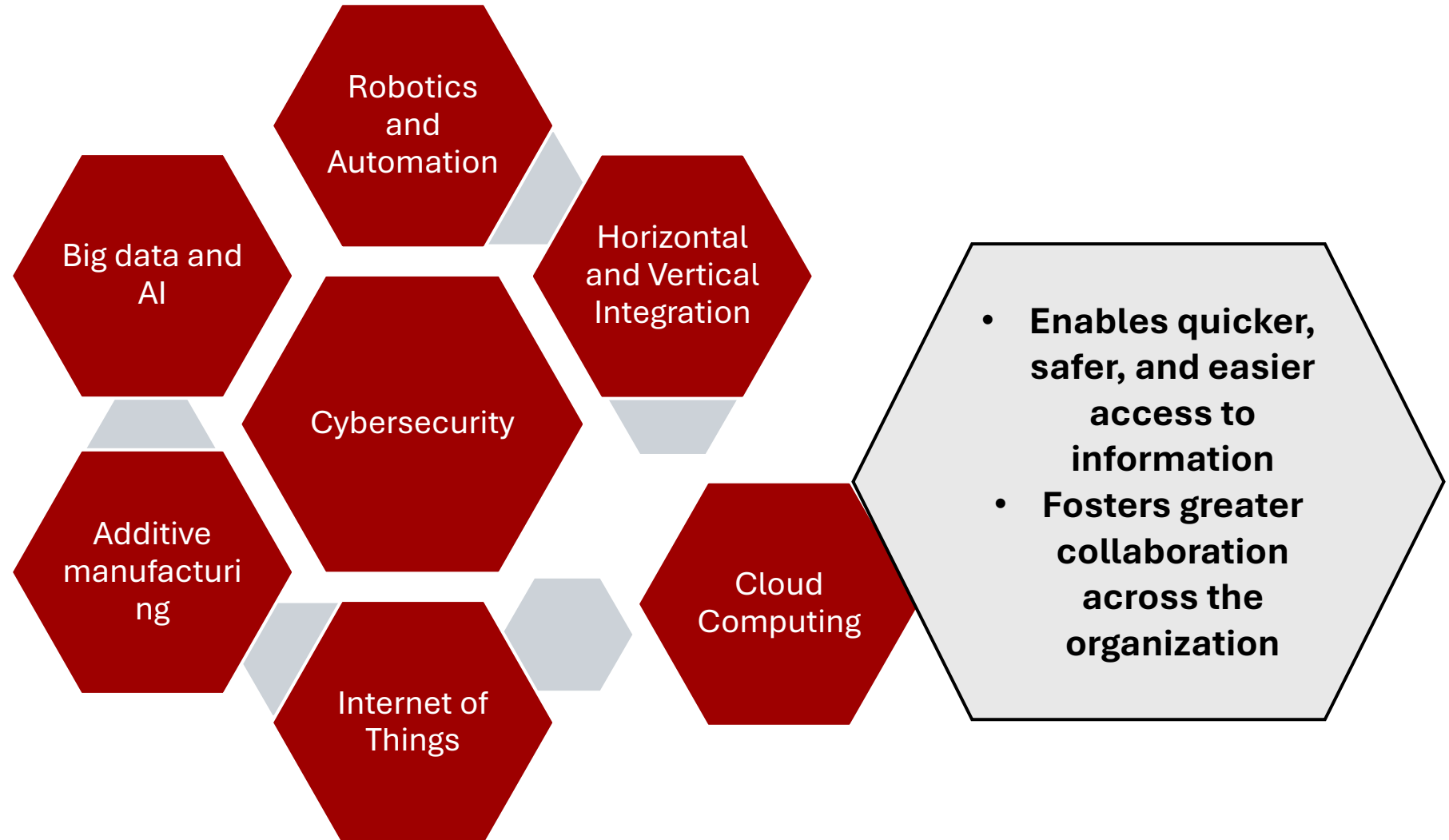
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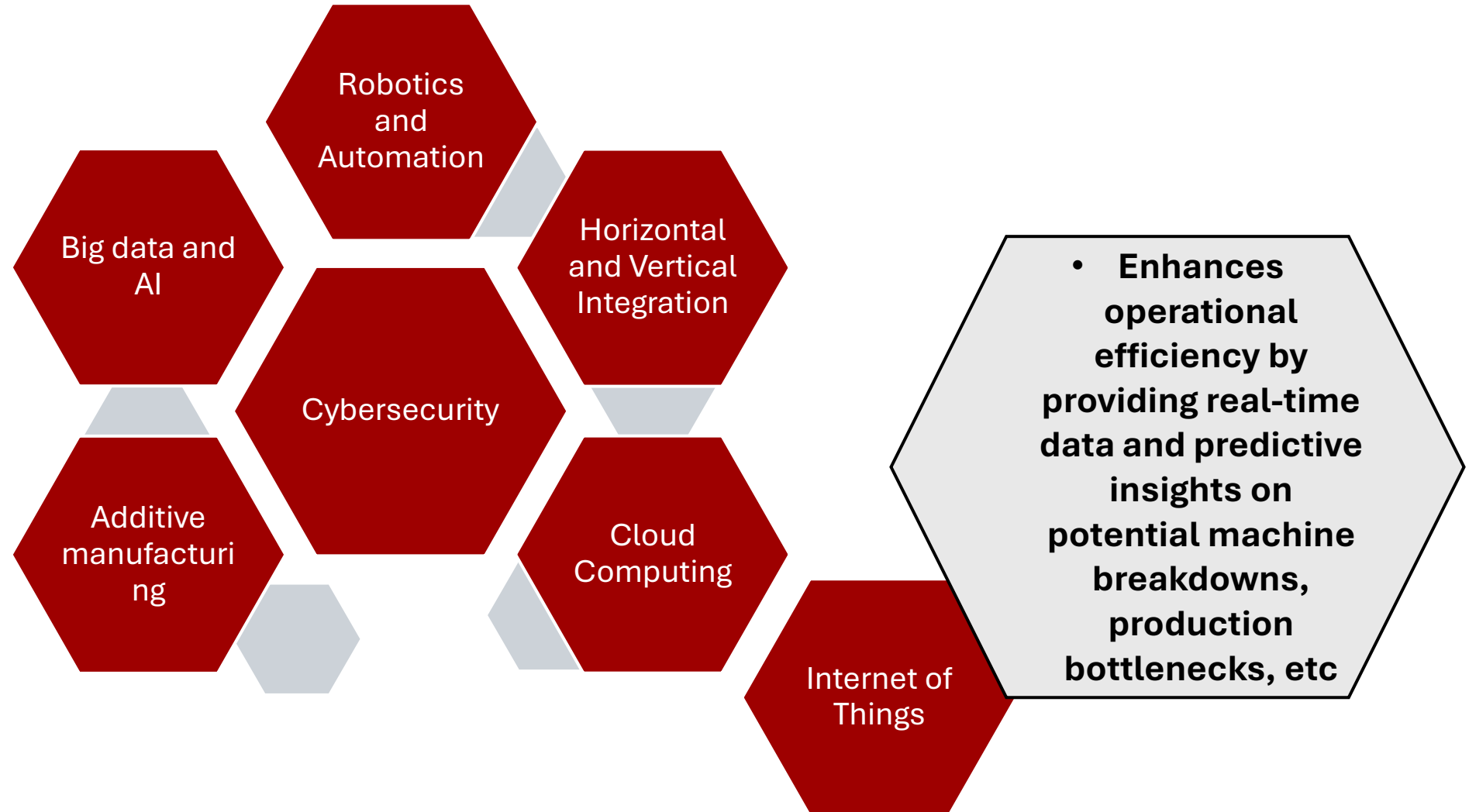
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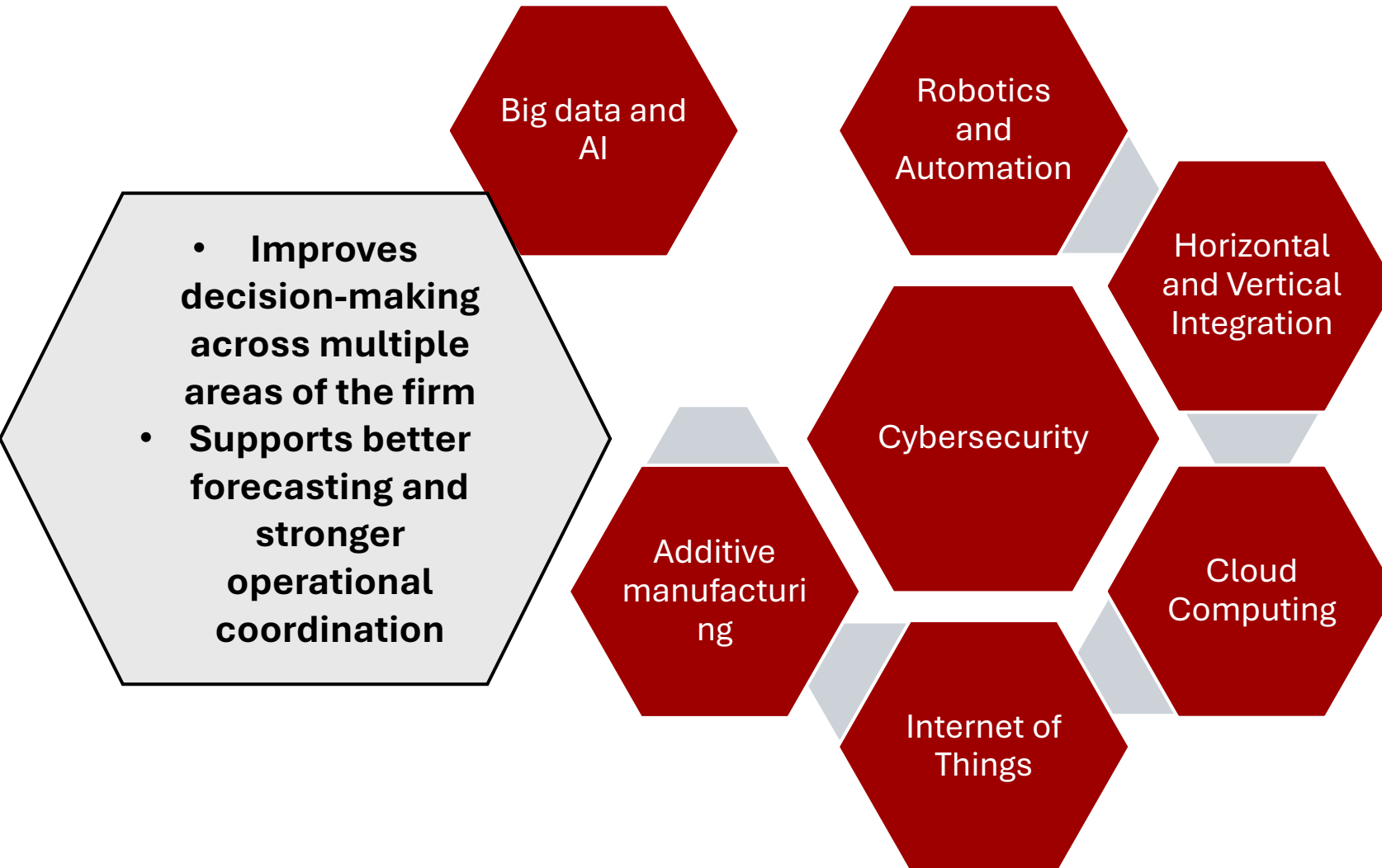
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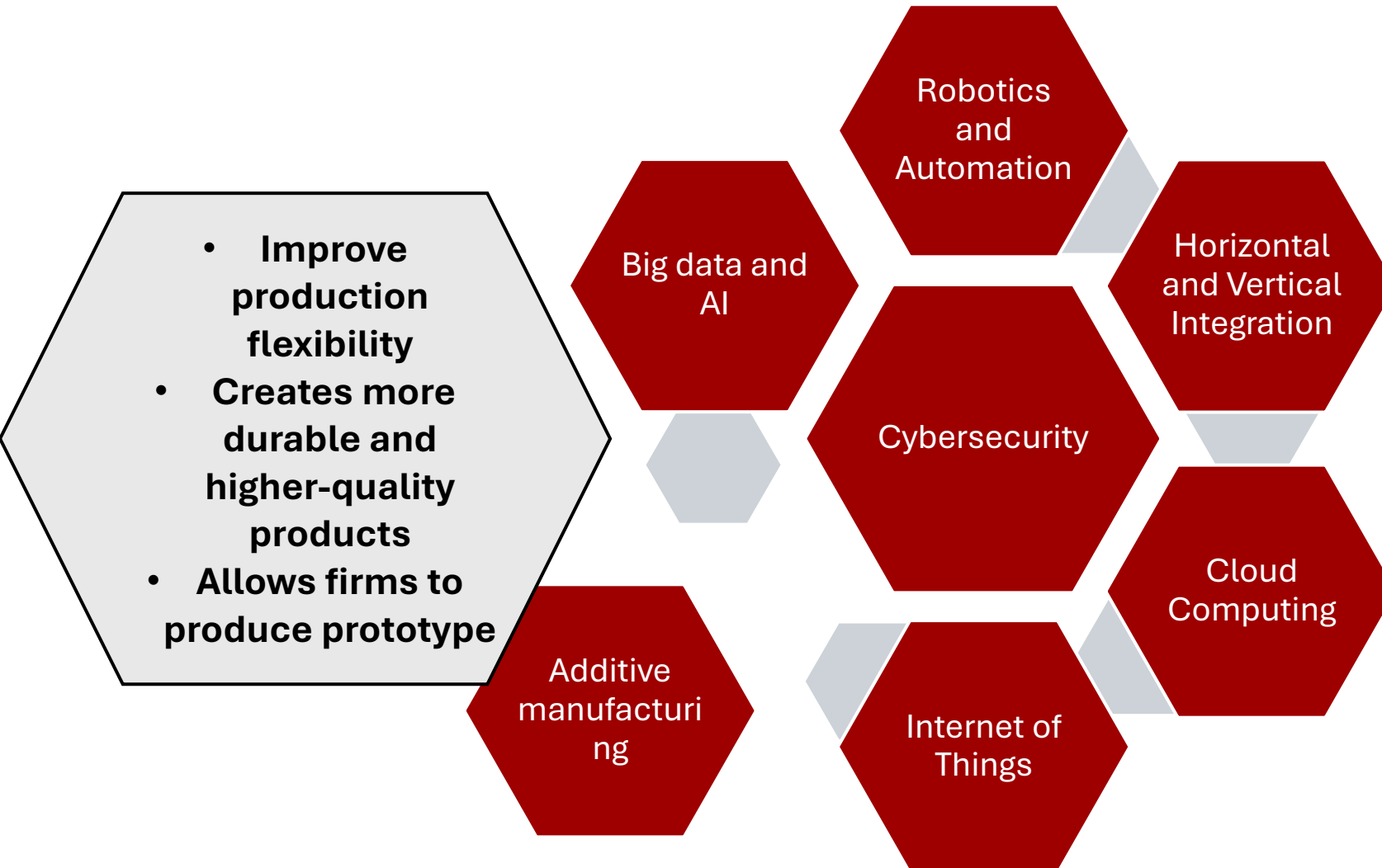
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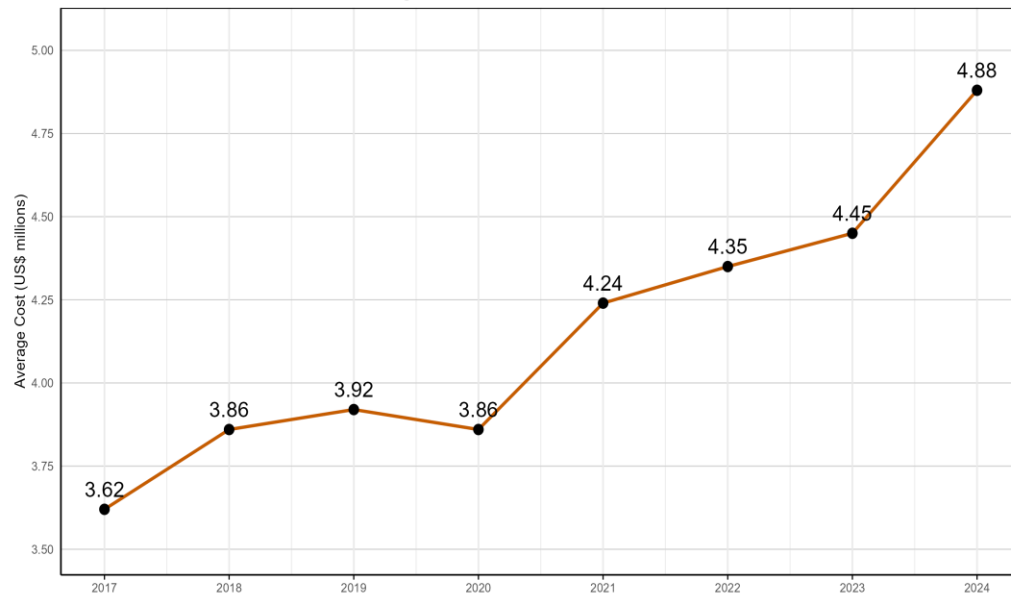
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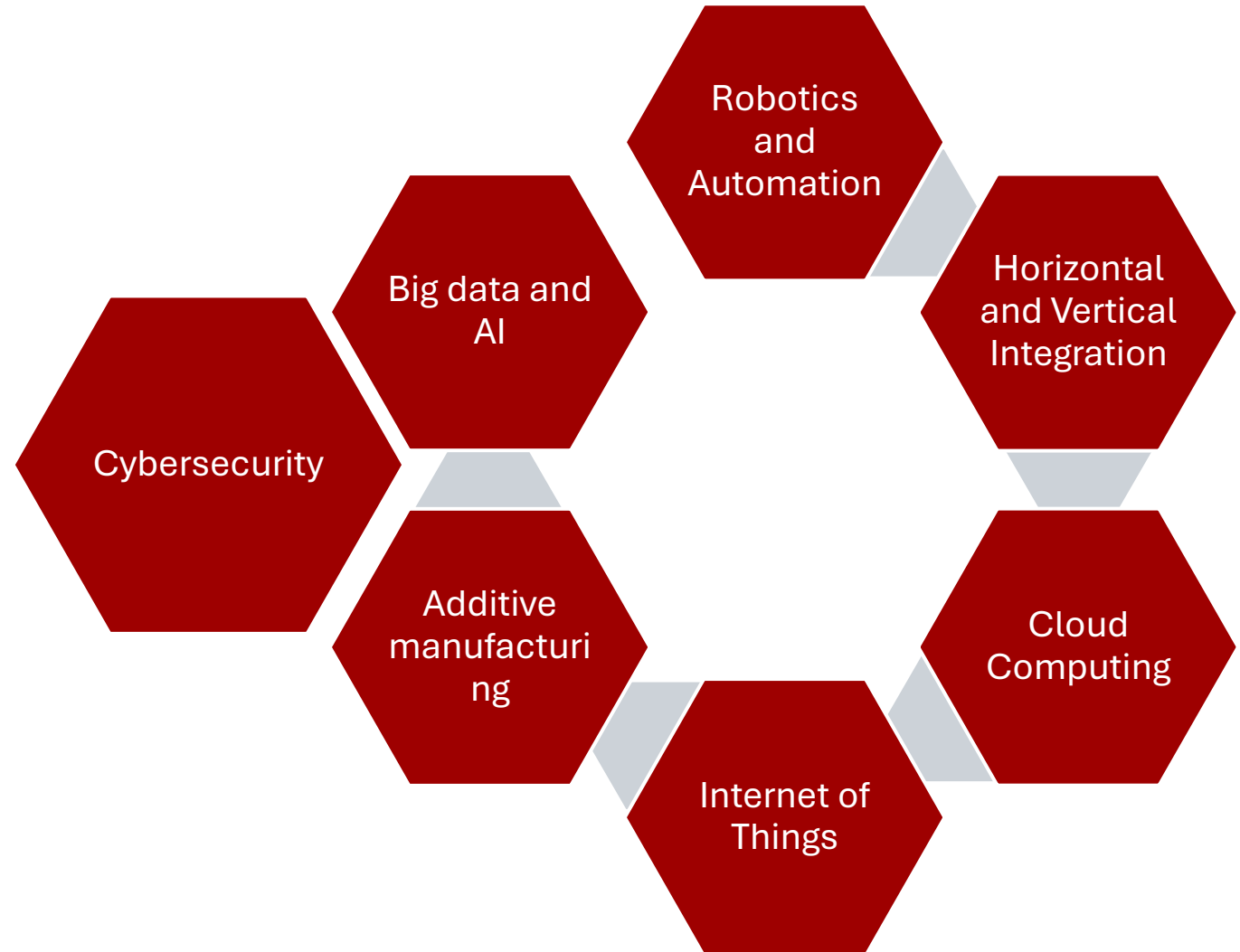
Cyber incidents' risks:

- Reputational Damage
- Supply Chain Disruptions
 - Erosion of trust
- Reduced incentives to invest

Global Average Cost of a Data Breach, 2017 to 2024



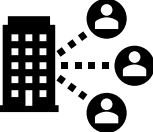
Source: World Bank (2022), Digitalizing SMEs to Boost Competitiveness. Based on data from IBM (2023, 2024).



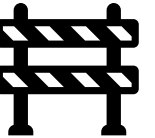
Policy Recommendation

Research and Development

Pro-Competitive Measures

1. Stronger competition raises labour productivity 
2. It pushes firms to innovate and invest in R&D 
3. Weak competition reduces frontier growth and innovation 

Interprovincial Trade

1. Internal trade rules remain fragmented across provinces 
2. These barriers create delays, costs, and lost productivity 
3. Fewer barriers would improve specialization and spillover gains 

Canada's current moment of economic self-reflection is an uncommon opportunity. There is an atypical appetite for risk and revision. We feel optimistic about the role and mandate of PrairesCan and its ability to coordinate with firms to deliver real productivity growth across the Prairies (and, in doing so, across Canada!)

Executive Summary

Proactive Control

Pain points

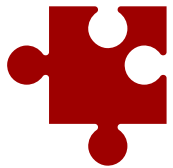


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Q&A Period

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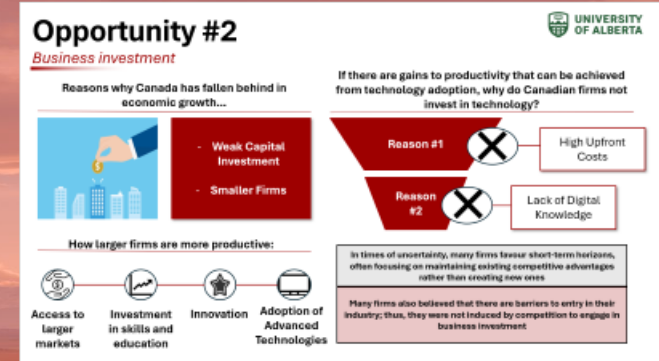
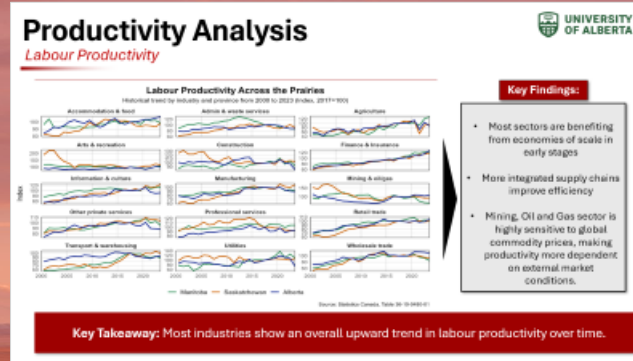


PrairiesCan

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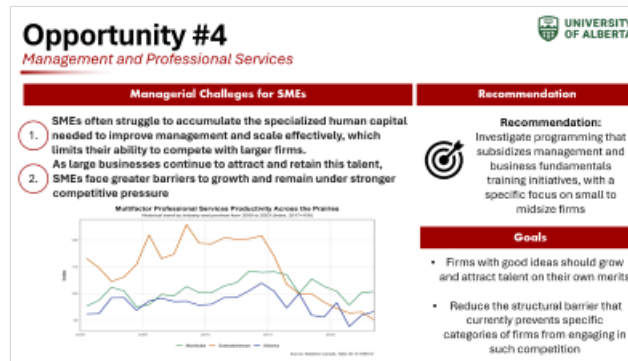




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